

Xiaoshan Wu

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Education

The University of Hong Kong 2023.09 – Present

Ph.D. in Electrical and Electronic Engineering, GPA: 4.0/4.3

Awards: Hong Kong PhD Fellowship, HKU Presidential Scholarship

Supervisors: Dr. Xiaojuan Qi, Dr. Zhongrui Wang

Zhejiang University 2019.09 – 2023.05

B.S. in Computer Engineering, GPA: 3.99/4.0

Honors: Outstanding Graduate in Zhejiang University

University of Illinois at Urbana-Champaign 2019.09 – 2023.05

B.S. in Computer Engineering, GPA: 3.75/4.0

Honors: Dean's List (2020, 2021)

Research Interest

My research focuses on high-resolution, temporally-consistent 3D/4D scene reconstruction and perception, particularly under challenging conditions such as dynamic motion, low light, or extreme illumination. I explore fusion of event-based and frame-based inputs, unified token representation, and video-spatial VLMs.

Work Experience

Research Intern, Megvii Inc. 2022.12 – 2023.03

Worked on depth-enhanced semantic segmentation for autonomous driving scenes.

Awards

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- Hong Kong PhD Fellowship (2023)
 - HKU Presidential Scholarship (2023)
 - Y S & Christabel Lung Postgraduate Scholarship (2023)
 - Dean's List, UIUC (2020, 2021)
 - Outstanding Student and Academic Excellence Award, ZJU (2020–2021)

Publications

Under Review

- **X. Wu**, Y. Yu, B. Wang, Y.-H. Huang, X. Lyu, B. Zhang, Z. Wang, X. Qi. *EAG3R: Event-Augmented 3D Geometry Estimation for Dynamic and Extreme-Lighting Scenes*. Under review at *NeurIPS* 2025.
- **X. Wu**, X. Lyu, B. Wang, Y. Yu, Z. Wang, X. Qi. *Segment Beyond Frames: Anytime Interframe Semantic Segmentation with Event Cameras*. Under review at *NeurIPS* 2025.

- X. Lyu, M. Liu, **X. Wu**, S. Shi, X. Qi. *Consistent Video Geometric Estimation with Stream Pack Tokens*. Under review at *SIGGRAPH Asia* 2025.

Journal and Conference Papers

1. **X. Wu**, W. He, M. Yao, Z. Zhang, Y. Wang, B. Xu, G. Li. *Event-based depth prediction with deep spiking neural network*. *IEEE Trans. Cogn. Dev. Syst.*, 2024. (*Under review)
2. H. Chen, J. Yang, J. Chen, S. Wang, D. Wang, X. Tian, Y. Yu, X. Chen, **X. Wu**, *et al.* *Continuous-time digital twin with analog memristive neural ordinary differential equation solver*. *Science Advances*, vol. 11, no. 22, eadr7571, 2025.
3. B. Wang, X. Zhang, S. Wang, N. Lin, Y. Li, Y. Yu, Y. Zhang, J. Yang, **X. Wu**, *et al.* *Topology optimization of random memristors for input-aware dynamic SNN*. *Science Advances*, vol. 11, no. 16, eads5340, 2025.
4. Y. Yu, S. Wang, M. Xu, W. Zhang, B. Wang, J. Yang, Y. Zhang, **X. Wu**, *et al.* *Random memristor-based dynamic graph CNN for efficient point cloud learning at the edge*. *npj Unconventional Computing*, vol. 1, no. 1, article 6, 2025.
5. C. Li, X. Chen, Z. Zhang, **X. Wu**, T. Yu, R. Bie, D. Yang, Y. Yao, Z. Wang, L. Sun. *Charge-selective 2D heterointerface-driven multifunctional floating gate memory for in situ sensing-memory-computing*. *Nano Letters*, vol. 24, no. 47, pp. 15025–15034, 2024.